



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

September
2006
Year 11 Yearly
Examination

Mathematics

(2 Unit Continuers)

General Instructions

- Reading Time – 5 Minutes
- Working time – 1 ½ Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- EACH SECTION IS TO BE RETURNED IN A SEPARATE BUNDLE. Section A (Questions 1 & 2) and Section B (Questions 3 & 4)
- All necessary working should be shown in every question.

Total Marks – 60

- Attempt questions 1 – 4

Examiner: *F Nesbitt*

SECTION A

QUESTION 1

- (a) Show that $\frac{1}{\sqrt{7}+2} - \frac{1}{\sqrt{7}-2}$ is rational. 2
- (b) Find $\lim_{x \rightarrow 5} \frac{x^2 - 5}{x - 5}$ 1
- (c) Differentiate:
- (i) $24x(2x^2 - 1)$ 2
- (ii) $\frac{x-5}{2x+3}$ 2
- (iii) $(7x^2 - 6x)^3$ 2
- (d) Simplify fully: $\log_3 \sqrt{27} - \log_2 \frac{1}{8} + \log_5 125$ 2
- (e) Solve the inequation $|3x - 1| \leq 5$ and graph the solution on a number line. 2
- (f) Solve for m : $2^{2m} - 3 \times 2^m - 10 = 0$ 2

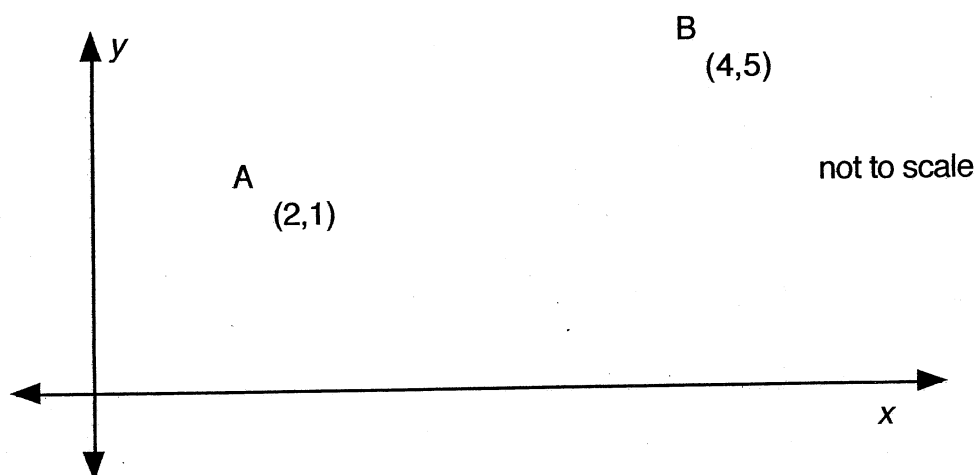
QUESTION 2

(a) For the parabola $(x+2)^2 = 8(y-1)$, find:

- (i) the coordinates of its vertex
- (ii) the coordinates its focus
- (ii) the equation of its directrix.

3

(b)



In the diagram above, The points A and B have coordinates (2,1) and (4,5).

- (i) Find M, the midpoint of AB. 1
- (ii) Find the Gradient of AB. 1
- (iii) Find the equation of the perpendicular bisector of AB. 2
- (iv) This perpendicular through M [in part (iii)] meets the y axis at D.
Find the coordinates of D 1
- (v) Find the perpendicular distance from D to AB 2
- (vi) Find the area of the triangle ADB 2

(c) Find from first principles, the gradient of the curve

$$x^2 + x \text{ at the point where } x=1$$

3

SECTION B - START A NEW BOOKLET

QUESTION 3

(a) For the curve $2x^3 + 3x^2 - 12x - 2$:

- (i) Find any turning points.
- (ii) Determine the nature of each turning point.
- (iii) Sketch the curve $-3 \leq x \leq 2$ showing all relevant features.
- (iv) Find the minimum value of the curve in the given domain.

6

(b) In the arithmetic series $2 + 5 + 8 + 11 + \dots$, find:

- (i) the 15th term.
- (ii) the sum of the first 15 terms.

4

(c) a, b and 10 form an arithmetic sequence.

- (i) Write an equation in a and b using the above information:

a, b and 18 form a geometric sequence.

- (ii) Write a second equation in a and b using the above information
- (iii) Solve the equations in (i) and (ii) to find 2 sets of values for a and b.

5

QUESTION 4

- (a) Find the value(s) of k if the roots of the equation $x^2 + kx + 36 = 0$ are real and distinct. 2
- (b) In an “organic” fruit shop only 60% of the oranges and 80% of the apples are really organic. George bought 2 apples and one orange. What is the probability that at least one of the three pieces of fruit was really organic? 2
- (c) A closed cylindrical can is made from 100π square centimetres of metal. If h is the height and r is the radius,
- (i) show that $h = \frac{50}{r} - r$ 2
- (ii) Hence find an expression for the volume in terms of r . 1
- (iii) Find the maximum possible volume of the can and show why it is a maximum. Answer to the nearest centimetre. 4
- (d) From a point O , an observer can see a lighthouse L on a bearing of 285° .
An oil rig R can also be seen by the observer on a bearing of 215° . The lighthouse is 12 km from the oil rig and on a bearing of 012° from the oil rig.
- (i) Find the distance, to the nearest kilometre, of the oil rig from the observer. 3
- (ii) Find the bearing of the oil rig from the lighthouse. 1

END OF PAPER

SECTION A

Question 1

$$(a) \frac{1}{(\sqrt{7}+2)} \times \frac{(\sqrt{7}-2)}{(\sqrt{7}-2)} - \frac{1}{(\sqrt{7}-2)} \times \frac{(\sqrt{7}+2)}{(\sqrt{7}+2)}$$

$$= \frac{\sqrt{7}-2}{7-4} - \frac{(\sqrt{7}+2)}{7-4}$$

$$= \frac{\sqrt{7}-2-\sqrt{7}-2}{3}$$

$$= -\frac{4}{3} \text{ which is rational}$$

$$\therefore \frac{1}{\sqrt{7}+2} - \frac{1}{\sqrt{7}-2} \text{ is rational}$$

$$(b) \lim_{x \rightarrow 5} \frac{x^2-5}{x-5}$$

we can't factorise x^2-5 into factors that will cancel with $x-5$.

Therefore, the limit does not exist $\left(\frac{20}{0}\right)$.

$$(c)(i) \text{ let } y = 24x(2x^2-1)$$

$$y = 48x^3 - 24x$$

$$y' = 144x^2 - 24$$

$$(ii) \text{ let } y = \frac{x-5}{2x+3} \quad \begin{matrix} v = 2x+3 \\ v' = 2 \end{matrix} \quad \begin{matrix} u = x-5 \\ u' = 1 \end{matrix}$$

$$y' = \frac{(2x+3) \cdot 1 - 2(x-5)}{(2x+3)^2}$$

$$y' = \frac{2x+3-2x+10}{(2x+3)^2}$$

$$y' = \frac{13}{(2x+3)^2}$$

$$(iii) \text{ let } y = (7x^2-6x)^3$$

$$y' = 3(7x^2-6x)^2(14x-6)$$

$$y' = 6(7x-3)(7x^2-6x)^2$$

$$\begin{aligned}
 (d) \quad & \log_3 \sqrt{27} - \log_2 \frac{1}{8} + \log_5 125 \\
 &= \log_3 3^{3/2} - \log_2 2^{-3} + \log_5 5^3 \\
 &= \frac{3}{2} \log_3 3 + 3 \log_2 2 + 3 \log_5 5 \\
 &\quad \text{(since } \log_a a = 1 \text{)} \\
 &= \frac{3}{2} + 3 + 3 \\
 &= \frac{15}{2}
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad & |3x-1| \leq 5 \\
 & 3x-1 \leq 5 \quad \text{or} \quad -(3x-1) \leq 5 \\
 & 3x \leq 6 \quad \quad \quad -3x+1 \leq 5 \\
 & x \leq 2 \quad \quad \quad -3x \leq 4 \\
 & \quad \quad \quad x \geq -\frac{4}{3}
 \end{aligned}$$



$$(f) \quad 2^{2m} - 3 \times 2^m - 10 = 0$$

$$(2^m)^2 - 3 \times 2^m - 10 = 0$$

$$\text{let } a = 2^m$$

$$a^2 - 3a - 10 = 0$$

$$(a-5)(a+2) = 0$$

$$a-5=0$$

$$a=5$$

$$a+2=0$$

$$a=-2$$

$$2^m = 5$$

$$m = \log_2 5$$

$$m \approx 2.322$$

$$\begin{array}{r}
 x \mid -10 \\
 + \quad -3 \\
 \hline
 -5, 2
 \end{array}$$

$$2^m = -2 \quad (\text{since } 2^m > 0)$$

$2^m = -2$ has no solution

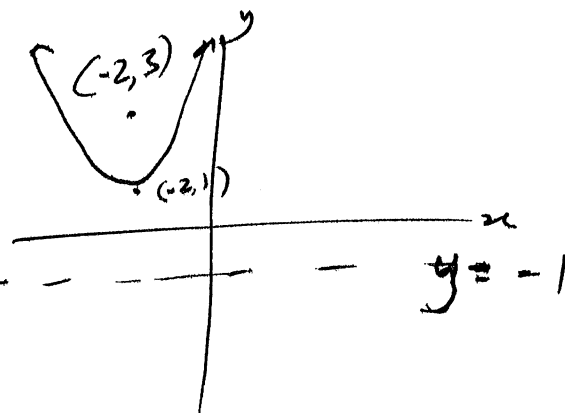
Question 2

(a) $(x+2)^2 = 8(y-1)$

is in the form

$$(x-h)^2 = 4a(y-k)$$

where (h, k) is vertex
 a is focal length



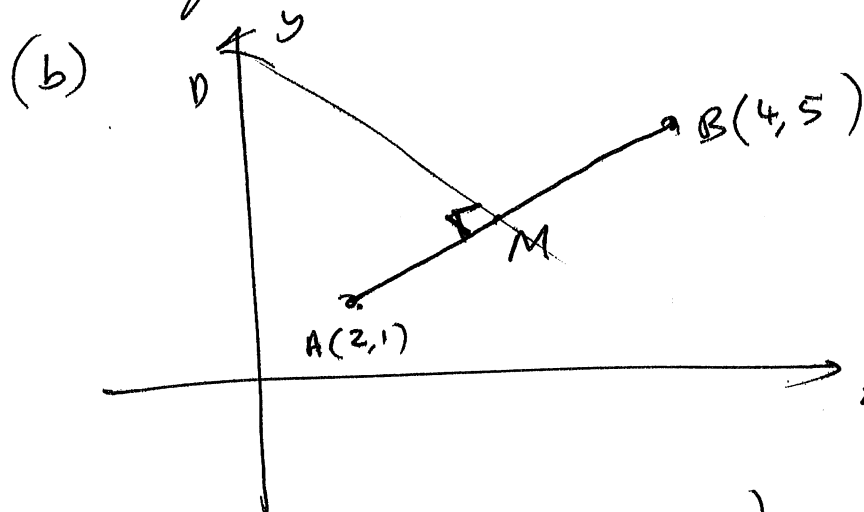
(i) vertex is $(-2, 1)$

(ii) $4a = 8$

$$a = 2$$

focus is $(-2, 3)$

(iii) equation of directrix is $y = -1$



(i) $M \left(\frac{2+4}{2}, \frac{1+5}{2} \right)$
 $M(3, 3)$

(ii) $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m_{AB} = \frac{5-1}{4-2}$$

$$m_{AB} = \frac{4}{2}$$

$$m_{AB} = 2$$

(iii) $m_2 = -\frac{1}{2}$ (perpendicular lines $m_1 \times m_2 = -1$)

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -\frac{1}{2}(x - 3)$$

$$2y - 6 = -x + 3$$

$$x + 2y - 9 = 0$$

(iv) let $x = 0$

$$2y - 9 = 0$$

$$2y = 9$$

$$y = \frac{9}{2}$$

$$\therefore D\left(0, \frac{9}{2}\right)$$

(v) The perpendicular distance from D to AB is the length DM.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$DM = \sqrt{(3 - 0)^2 + \left(3 - \frac{9}{2}\right)^2}$$

$$DM = \sqrt{9 + \frac{9}{4}}$$

$$DM = \sqrt{\frac{45}{4}}$$

$$DM = \frac{3\sqrt{5}}{2}$$

(vi) $AB = \sqrt{(4 - 2)^2 + (5 - 1)^2}$

$$AB = \sqrt{4 + 16}$$

$$AB = \sqrt{20}$$

$$AB = 2\sqrt{5}$$

$$\text{Area} = \frac{1}{2} \times AB \times DM = \frac{1}{2} \times 2\sqrt{5} \times \frac{3\sqrt{5}}{2} = \frac{15}{2} \text{ units}^2$$

$$(c) \quad y = x^2 + x$$

$$\text{let } f(x) = x^2 + x$$

$$f(x+h) = (x+h)^2 + (x+h)$$

$$= x^2 + 2xh + h^2 + x + h$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2xh + \cancel{h^2} + \cancel{x} + h - (\cancel{x^2} + \cancel{x})}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h} (2x + h + 1)}{\cancel{h}}$$

$$= 2x + (0) + 1$$

$$= 2x + 1$$

$$\therefore y' = 2x + 1 \quad \text{when } x = 1$$

$$m_T = 2(1) + 1$$

$$= 3$$

2006 Mathematics Continuers 2-Unit Yearly: **Section B** solutions

3. (a) For the curve $y = 2x^3 + 3x^2 - 12x - 2$:

(i) Find any turning points.

2

Solution: $\frac{dy}{dx} = 6x^2 + 6x - 12,$
 $= 6(x^2 + x - 2),$
 $= 6(x + 2)(x - 1),$
 $= 0 \text{ when } x = -2, 1.$
 $\therefore$ Turning points are at $(-2, 18)$ and $(1, -9)$.

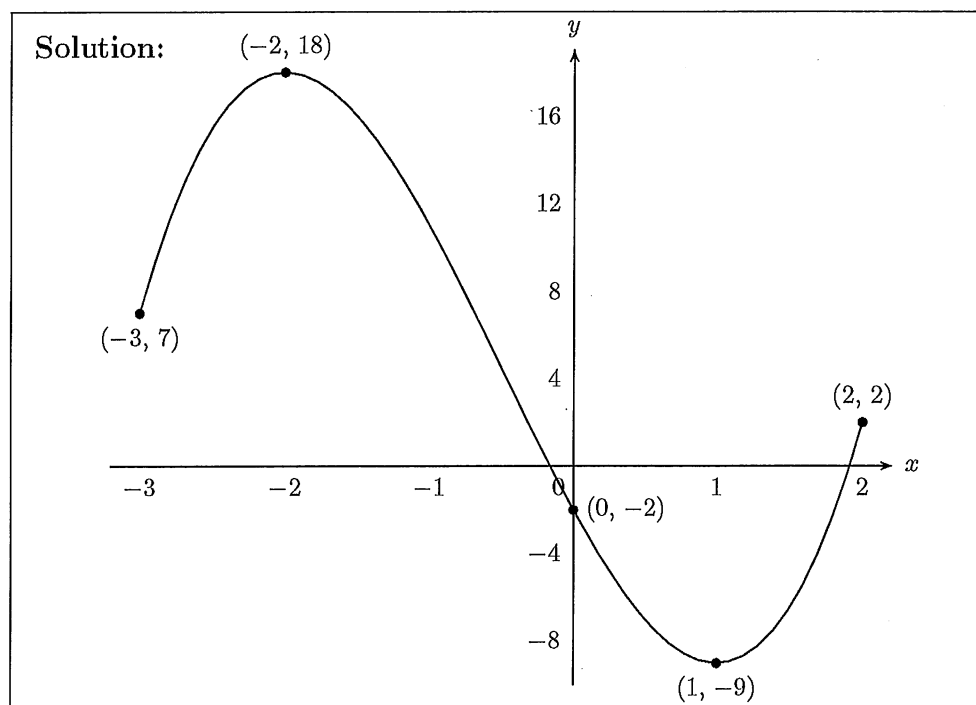
(ii) Determine the nature of each turning point.

1

Solution: $\frac{d^2y}{dx^2} = 12x + 6,$
 $= -18 \text{ when } x = -2,$
 $= 18 \text{ when } x = 1.$
 $\therefore$ Maximum at $(-2, 18)$ and minimum at $(1, -9)$.

(iii) Sketch the curve $-3 \leq x \leq 2$ showing all relevant features.

2



(iv) Find the minimum value of the curve in the given domain.

1

Solution: The minimum value is -9 .

(b) In the arithmetic series $2 + 5 + 8 + 11 + \dots$, find:

(i) the 15th term,

2

$$\begin{aligned}\text{Solution: } a &= 2, \quad d = 3. \\ \therefore U_{15} &= 2 + (15 - 1)3, \\ &= 44.\end{aligned}$$

(ii) the sum of the first 15 terms.

2

$$\begin{aligned}\text{Solution: } S_{15} &= \frac{15}{2}(2 + 44), \\ &= 345.\end{aligned}$$

(c) a , b , and 10 form an arithmetic sequence.

(i) Write an equation in a and b using the above information.

1

$$\begin{aligned}\text{Solution: } 10 - b &= b - a, \\ a &= 2b - 10.\end{aligned}$$

a , b , and 18 form a geometric sequence.

(ii) Write a second equation in a and b using the above information.

1

$$\begin{aligned}\text{Solution: } \frac{18}{b} &= \frac{b}{a}, \\ b^2 &= 18a.\end{aligned}$$

(iii) Solve the equations in (i) and (ii) to find 2 sets of values for a and b .

3

$$\begin{aligned}\text{Solution: } b^2 &= 18(2b - 10), \\ b^2 - 36b + 180 &= 0, \\ (b - 6)(b - 30) &= 0. \\ \therefore b &= 6 \text{ or } 30, \\ a &= 2 \text{ or } 50.\end{aligned}$$

4. (a) Find the value(s) of k if the roots of the equation $x^2 + kx + 36 = 0$ are real and distinct. 2

Solution:

$$\begin{aligned}\Delta &> 0, \\ \text{i.e., } k^2 - 4 \times 36 &> 0, \\ k^2 &> 12^2, \\ \therefore k &< -12, \quad k > 12.\end{aligned}$$

- (b) In an “organic” fruit shop only 60% of the oranges and 80% of the apples are really organic. George bought two apples and one orange. What is the probability that at least one of the three pieces of fruit was really organic? 2

Solution: $P(\text{at least one}) = P(\text{not none}),$

$$\begin{aligned}&= 1 - \frac{1}{5} \times \frac{1}{5} \times \frac{2}{5}, \\ &= \frac{123}{125}.\end{aligned}$$

- (c) A closed cylindrical can is made from 100π square centimetres of metal. If h is the height and r is the radius,

- (i) show that $h = \frac{50}{r} - r$. 2

Solution: Surface area, $100\pi = 2\pi r^2 + 2\pi rh,$

$$\begin{aligned}50 &= r^2 + rh, \\ rh &= 50 - r^2, \\ h &= \frac{50}{r} - r.\end{aligned}$$

- (ii) Hence find an expression for the volume in terms of r . 1

Solution: Volume, $V = \pi r^2 h,$

$$\begin{aligned}&= \pi r^2 \left(\frac{50}{r} - r \right), \\ &= (50\pi r - \pi r^3), \text{ or} \\ &= (50 - r^2)\pi r.\end{aligned}$$

- (iii) Find the maximum possible volume of the can and show why it is a maximum.
Answer to the nearest cubic centimetre.

4

Solution:

$$\begin{aligned}\frac{dV}{dr} &= (50 - 3r^2)\pi, \\ &= 0 \text{ when } r^2 = \frac{50}{3},\end{aligned}$$

$$\text{i.e. when } r = 5\sqrt{\frac{2}{3}} \text{ cm.}$$

$$\begin{aligned}\frac{d^2V}{dr^2} &= -6\pi r, \\ &< 0 \text{ when } r > 0.\end{aligned}$$

$$\therefore \text{Max. volume when } r^2 = \frac{50}{3} \text{ cm.}$$

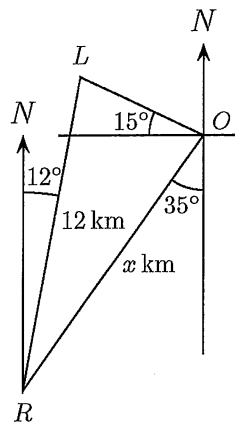
$$\begin{aligned}\text{Max. volume} &= \left(50 - \frac{50}{3}\right) \times 5\sqrt{\frac{2}{3}} \times \pi \text{ cm}^3, \\ &\approx 427.516609974 \text{ cm}^3 \text{ by calculator,} \\ &\approx 428 \text{ cm}^3.\end{aligned}$$

- (d) From a point O , an observer can see a lighthouse L on a bearing of 285°T . An oil rig R can also be seen by the observer on a bearing of 215°T . The lighthouse is 12 km from the oil rig and on a bearing of 012°T from the oil rig.

- (i) Find the distance, to the nearest kilometre, of the oil rig from the observer.

3

Solution:



$$\begin{aligned}\angle LOR &= (15 + 90 - 35)^\circ, \\ &= 70^\circ.\end{aligned}$$

$$\begin{aligned}\angle ORL &= (35 - 12)^\circ, \\ &= 23^\circ.\end{aligned}$$

$$\begin{aligned}\angle RLO &= (180 - 70 - 23)^\circ, \\ &= 87^\circ.\end{aligned}$$

$$\begin{aligned}\frac{x}{\sin 87^\circ} &= \frac{12}{\sin 70^\circ}, \\ x &= \frac{12 \sin 87^\circ}{\sin 70^\circ},\end{aligned}$$

$$\approx 12.75263225 \text{ by calculator.}$$

$\therefore$ Rig is about 13 km from the observer.

- (ii) Find the bearing of the oil rig from the lighthouse.

1

$$\begin{aligned}\text{Solution: Bearing} &= 12^\circ + 180^\circ, \\ &= 192^\circ\text{T}.\end{aligned}$$